

Telescoping Network Reinstatement in Self-Aware Networks

Dendritic Three-Dimensional Memory, Oscillatory Winner-Take-Most Selection, and NAPOT Reconstruction

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Abstract

This paper develops a Self-Aware Networks (SAN) hypothesis about how a learned state in one dendritic arbor can help regenerate a much larger three-dimensional perceptual or mnemonic configuration. The proposal combines nonlinear dendritic pattern recognition, oscillatory winner-take-most competition, recurrent excitation and inhibition, and Neural Array Projection Oscillation Tomography (NAPOT). During learning, a distributed event changes synapses, dendritic compartments, cellular excitability, and circuit relationships. In the stronger SAN interpretation, a dendritic arrangement can store a three-dimensional relational projection of the network neighborhood that surrounded the neuron during the event. During retrieval, a partial cue activates that learned disposition. Prepared neurons or assemblies reach threshold earlier during an oscillatory opportunity window, recruit local inhibitory circuitry, and bias which sparse population survives the cycle. Their outputs then propagate through recurrent and long-range pathways, regenerating an approximation of the earlier three-dimensional state while incorporating current sensory, bodily, and action context.

“Storage of a 3D image” is used here in a physically testable sense. The dendrite may store a structured projection, basis component, or addressable transformation capable of contributing to reconstruction; it need not contain a voxel-for-voxel miniature watched by an inner observer. The experienced image is the distributed network’s active observer-action state. The model extends, but is not identical to, Hierarchical Temporal Memory (HTM), sparse distributed representations (SDRs), hippocampal pattern completion, or gamma-cycle competition. For a declared finite-dimensional scene family, a full-column-rank dendritic measurement operator gives exact noiseless reconstruction and a norm bound under additive noise. A deterministic fixture reconstructed a 1,728-voxel structured volume from 24 measurements over a six-dimensional basis with relative error $9.71\text{e-}16$, satisfied its noise bound, and selected the correct stored perspective. A second deterministic fixture learned the 24-by-6 forward operator from 2,400 noisy examples and reconstructed 384 held-out scenes through a stable excitatory-minus-inhibitory recurrence without a pseudoinverse at inference. Its median relative error was 0.0561, compared with 0.6563 after one iteration and 1.4075 after measurement-scene pairing was shuffled. A third fixture imposed nonnegative excitatory outputs, subtractive inhibitory outputs, bounded states, delayed conductance-style recurrence, projected local updates, and a frozen multi-condition support rule. It reached median error 0.4915 and prototype accuracy 0.8594, but did not improve accuracy over matched feedforward inference and did not meet the required error advantage over the no-inhibition ablation. A signed ridge comparator was also better (median error 0.3745; accuracy 0.9938). The primary constrained-recurrence hypothesis was therefore not supported. A separately frozen 12-seed sensitivity audit produced zero full primary passes: no seed met the required accuracy gain over feedforward or error gain over no inhibition, while every seed beat shuffled pairing and retained sign / state integrity. The constrained fixture uses supervised target-code errors, evaluates new noisy samples from the same eight prototype family, reconstructs six latent coordinates rather than a learned voxel volume, and computes prototype accuracy through an external nearest-prototype decoder. These are constructive sufficiency and pipeline results for the declared models, not evidence that biological dendrites use the mechanism. Alternative models and multiscale perturbation-and-rescue tests are provided.

Keywords: Self-Aware Networks; SAN; NAPOT; dendritic computation; three-dimensional memory;

winner-take-all; winner-take-most; inhibition; sparse distributed representation; pattern completion; tonic canvas; phasic ink; HTM

Scope and evidence boundary

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It proves representational sufficiency only for a declared finite-dimensional model, reports deterministic synthetic reconstruction, and records a predeclared constrained-recurrence test whose primary hypothesis was not supported in its original seed or any of 12 fixed-hyperparameter sensitivity seeds. All 18 references have been checked against their sentence-level roles. Independent specialist review remains welcome and may motivate a later version, but no unresolved local scientific, formatting, or release-package defect is being deferred to that review. This paper does not establish that biological dendrites store three-dimensional images, that NAPOT is implemented by brains, or that the simulated state is conscious.

1. The central problem

Memory research shows that learning changes synapses, dendritic spines, intrinsic excitability, molecular organization, and ensemble allocation. Distinct experiments demonstrate temporally linked ensemble overlap, latent engram access after induced amnesia, retrieval-dependent reconsolidation, and molecular maintenance of persistent memory (Cai et al., 2016; Ryan et al., 2015; Nader et al., 2000; Tsokas et al., 2024). Single-neuron stimulation can also bias behavior under controlled conditions (Houweling and Brecht, 2008). None of those findings establishes that one synapse, molecule, cell, or region contains a complete memory. The missing bridge is **causal expansion**: how can a local, history-dependent structure call back a much larger distributed state?

SAN answers through **changed readiness**. A past event changes what a receiver does when a later condition arrives. The local state need not resemble a camera image. It must alter a later receive-integrate-project relation.

1.1 Four claim classes

The argument is easier to evaluate when four claim classes remain separate.

1. **Established component science.** Dendrites compute nonlinearly; learning alters synaptic, dendritic, cellular, and ensemble state; recurrent circuits support pattern completion; excitation can recruit inhibition; and oscillatory circuit dynamics can affect spike timing and selection. These findings motivate components but do not establish the joined SAN mechanism.
2. **Constructive mathematical result.** Equations 15-16 prove exact recovery and a noise bound for a declared finite-dimensional scene model when its effective measurement matrix has full column rank. This is a theorem about the model, not a biological theorem.
3. **Synthetic computational result.** The pinned fixture demonstrates exact recovery, bounded noisy recovery, partial-cue perspective selection, and rank-preserving dropout. It verifies code and construction, not neural implementation.
4. **SAN / NAPOT hypothesis.** A learned dendritic projection may seed oscillatory winner-take-most selection and recurrent reinstatement of a larger three-dimensional, action-relevant network state. This class remains unconfirmed and is the target of the proposed experiments.

Chronology establishes that the joined hypothesis developed within SAN; it cannot move a statement from Class 4 into Classes 1-3.

The stronger NAPOT hypothesis is:

A dendritic arbor can store a learned three-dimensional relational projection of a prior network neighborhood. When a partial cue activates that disposition, the resulting neuronal output can seed an excitatory-inhibitory cascade that regenerates the larger scene or memory as a distributed network state.

This is **telescoping network reinstatement**. During learning, a large event folds into selected local structures. During retrieval, those structures unfold through the network. The process is lossy, context-sensitive, and revisable.

2. What it means for a dendrite to store a 3D image

Three distinct claims must not be confused.

2.1 Structural projection storage

A dendritic tree occupies three-dimensional space and receives pathway-specific inputs at particular branches and spines. Learning can change synaptic efficacy, receptor content, spine structure, branch excitability, and local nonlinear interactions. A resulting dendritic configuration can encode relations among input sources. In SAN, this is a candidate **physical projection** of the functional neighborhood surrounding the neuron during learning.

Calling this a stored 3D image means that the arrangement preserves enough spatial-relational and temporal structure to participate in later reconstruction. It does not require a literal photograph on the dendrite.

2.2 Regenerative or holographic storage

A local trace may be only a partial projection while still being sufficient to regenerate a larger pattern when embedded in the trained network. This resembles content-addressable or holographic reconstruction: a fragment can call back a distributed state because the network supplies the missing relations.

The dendrite therefore may store a 3D image in the same functional sense that a holographic fragment stores access to a larger image. The complete reconstruction depends on illumination and the rest of the optical system; analogously, dendritic reconstruction depends on current input, tonic state, recurrent circuitry, and action.

2.3 Active phenomenal reconstruction

The conscious or usable 3D scene is not independently present inside one branch. It is the active distributed state produced when cellular projections are integrated and propagated across the network. NAPOT names the proposed many-to-one-to-many reconstruction process. The network itself is the observer; there is no smaller viewer inspecting a dendritic picture.

These levels allow the phrase “a dendrite stores a 3D image” without either erasing the SAN claim or introducing a homunculus.

3. Source lineage inside SAN

The proposal continues the fractal and projection ideas recorded in Micah Blumberg’s early SAN materials, including [Intro.Fractal](#), the legacy SAN [Whitepaper](#), the dedicated [NAPOT page](#), and the committed

[whitepaperdraft3.md](#).

The legacy Whitepaper explicitly names **FRACTAL Conscious Perception: Functional Recursive Activity Cortical Telescoping Asymmetric Lensing**. It describes small synaptically stored patterns becoming differentiated circuit- and brain-scale activity. The NAPOT page separately states that NAPOT concerns not only perception but also how memories scale from synaptic connections to the whole brain, and it connects synaptically captured patterns, possible dimensionality gain during activation and transmission, and volumetric reconstruction. These pages are direct conceptual predecessors of telescoping network reinstatement.

An earlier public communication-protocol bridge appears in the April 2017 Neural Lace Podcast. Blumberg asks whether brain communication is more like TCP, with connection and feedback, or UDP, with connectionless transmission, and proposes **Alternating Transmission Protocol (ATP)** for a system that can alternate between those modes. The acronym is expressly paired with adenosine triphosphate, the cellular energy currency, to connect communication and metabolic expenditure. This is a conceptual ancestor of later PWD and NAPOT work: it establishes the receive / feedback / broadcast problem and the intentional ATP double meaning, but it does not retroactively contain the later phase-wave or tomographic mechanisms. See the immutable [Neural Lace Podcast transcript](#).

Four public 2022 video explanations provide a spoken source layer between those notes and the present formalization:

- [The Self Aware Networks Institute for Neurophysics, Artificial Neurology & Bio-Synthetic Interfaces \(July 28, 2022\)](#) identifies synapse-to-whole-brain memory extraction and NAPOT as connected research problems.
- [Self Aware Networks: A longer explanation of my research, my past work, and what I have built \(July 30, 2022\)](#) describes synaptic memory being magnified into whole-brain activity and frames the observer as an entified neural array rather than a hidden inner person.
- [One-hour explanation of NAPOT as the central thesis of SAN \(July 31, 2022\)](#) connects memory scaling, selective reception and re-expression, oscillatory coordination, neural rendering, 3D reconstruction, and distributed observation in one architecture.
- [Neural Array Projection Oscillation Tomography Theory, fifth revision \(September 18, 2022\)](#) develops the multiscale receiver-transmitter, tonic / phasic, inhibition, rendering, and observer-action account.

These first-party sources establish conceptual provenance, not medical validation. Automatic-transcript wording must be checked against video timestamps before quotation. Their physiological proposals are evaluated separately against primary literature.

This lineage also fixes the observer claim. The legacy pages and videos do not insert a smaller viewer behind the reconstruction. They assign partial reception, transformation, and re-expression to neural arrays and oscillating groups, whose coordinated activity collectively constitutes the observer-action process.

Several dated SVGN articles make the 3D reconstruction lineage more explicit:

- [Getting 3D Shapes for VR from 2D Images with AI to Make a Self-Aware Robot's Brain \(2017\)](#) proposes constructing an internal 3D space from image input, linking 2D words and sounds to 3D spatial concepts, cycling through internal 3D representations, and transferring broad, potentially lossy representations between neuron bundles through brainwaves.
- [Humans Are Metal Robots in a Valid Sense \(2018\)](#) supplies the embodied substrate framing developed around neural ions, temporal-spatial patterns, and physically implemented computation.
- [Addressing Criticism for "Humans Are Metal Robots" \(2018\)](#) states that perception is a rendered concept distributed through a fractal of feedback loops, emphasizes coincidence detection and subthreshold

dendritic computation, and summarizes the design intuition as a neural network combined with a three-dimensional point cloud and recursive feedback.

- [Analog Biological Backpropagation \(2024\)](#) extends the SAN lineage toward physically represented derivatives, loss, and biologically implemented credit assignment. In the present model, action outcomes can therefore modify which dendritic projections become easier to reconstruct in later cycles.

Together these sources show that 2D-to-3D reconstruction, embodied implementation, distributed feedback rendering, and action-dependent learning were components of SAN’s development before this paper’s present synthesis. They establish conceptual provenance; they do not by themselves validate the biological mechanism.

The present formulation adds a sharper causal sequence:

$$\begin{aligned}
 &\text{distributed 3D state} \rightarrow \text{dendritic projection} \\
 &\quad \rightarrow \text{timing advantage} \\
 &\quad \rightarrow \text{inhibitory selection} \\
 &\quad \rightarrow \text{network reinstatement.}
 \end{aligned} \tag{1}$$

4. Biological foundations

4.1 Nonlinear dendritic recognition

Dendrites are active computational structures. Spatially and temporally organized inputs can interact through NMDA, calcium, sodium, and voltage-dependent conductances. In CA1 pyramidal neurons, synchronous clustered input can recruit supralinear dendritic integration while dispersed asynchronous input is integrated differently (Gasparini and Magee, 2006). Nonlinear dendritic integration also has established mnemonic computational roles in hippocampal circuit models and evidence (Kaifosh and Losonczy, 2016). Learning can alter this computation through synaptic efficacy, morphology, receptor trafficking, and excitability.

This supports a pattern-sensitive local state. It does not establish a complete 3D memory code.

4.2 Predictive depolarization and earlier firing

Hawkins and Ahmad’s HTM sequence-memory model assigns contextual patterns to active dendritic segments. Contextual recognition depolarizes a predicted cell without necessarily causing a spike. When matching feedforward input arrives, the prepared cell spikes sooner than unprepared competitors. Fast local inhibition biases the sparse active population toward the prediction.

This gives changed readiness a concrete consequence: it becomes relative spike timing.

4.3 Excitation recruits inhibition

Most cortical pyramidal neurons are excitatory. Their output can excite other principal cells and recruit inhibitory interneurons. The recruited circuit can generate feedforward, feedback, lateral, or surround inhibition. Inhibitory neurons directly suppress their targets. A pyramidal neuron does not ordinarily release both glutamate and GABA; its excitatory output recruits a network that produces both excitation and inhibition. A recent dentate-gyrus experiment further shows that PV and SST interneuron perturbations can modulate network-immanent winner-take-all dynamics in freely behaving mice, with cell-type-specific effects (Hainmueller et al., 2026).

4.4 Oscillatory opportunity windows

Pyramidal-interneuron gamma and interneuron-network gamma mechanisms can create repeating windows of excitation and inhibition (Buzsáki and Wang, 2012). As inhibition decays, neurons again approach threshold. Learned dendritic readiness can give one cell or assembly a small phase or timing advantage. The first sufficiently prepared cells recruit inhibition that narrows the winner set.

The mechanism need not be gamma in every region. Oscillatory selection windows can occur through different frequencies, cell types, and circuit architectures.

5. Winner-take-most selection

Living circuits rarely implement a perfect single winner. The selected unit may be an assembly. Some competitors remain active. Inhibition can normalize, sharpen timing, produce surround suppression, synchronize, or disinhibit. This paper therefore uses **winner-take-most (WTM)**.

Let neuron i have tonic readiness $b_i(t)$, phasic input $p_i(t)$, dendritic memory match $d_i(t)$, and inhibitory drive $h_i(t)$:

$$u_i(t) = b_i(t) + p_i(t) + d_i(t) - h_i(t). \quad (2)$$

The first threshold-crossing time within opportunity window W_k is

$$\tau_i^{(k)} = \inf\{t \in W_k : u_i(t) \geq \theta_i(t)\}. \quad (3)$$

The selected set is

$$S_k = \left\{ i : \tau_i^{(k)} \leq \min_j \tau_j^{(k)} + \varepsilon_k \right\}. \quad (4)$$

A nonzero ε_k permits a near-synchronous assembly instead of one winner.

$$h_i(t) = \sum_{j \in S_k} w_{ij}^{EI} (\kappa_I * s_j)(t) + \sum_{\ell} w_{i\ell}^{II} (\kappa_{II} * s_\ell)(t). \quad (5)$$

$$A_k = \Phi(S_k, \mathbf{h}, \mathbf{W}^{EE}, \mathbf{W}^{EI}, \mathbf{W}^{IE}, \mathbf{c}_t). \quad (6)$$

where $\mathbf{c}_t$ is current sensory, bodily, motivational, and recurrent context. Selection is never determined by feedforward input alone.

6. Dendritic projection and NAPOT reconstruction

$$\mathbf{m}_i(\tau^+) = \mathcal{L}_i[\mathbf{X}_{\mathcal{N}(i)}(\tau), \mathbf{x}_i(\tau), \mathbf{a}_\tau, \mathbf{o}_\tau]. \quad (7)$$

Here $\mathbf{m}_i$ is the learned dendritic state and $\mathcal{N}(i)$ is the neuron's functional neighborhood. $\mathbf{a}_\tau$ and $\mathbf{o}_\tau$ denote action and outcome.

$$\mathbf{m}_i = \mathcal{P}_i[\mathbf{X}_{\mathcal{N}(i)}] + \boldsymbol{\eta}_i. \quad (8)$$

where $\mathcal{P}_i$ is a learned, nonlinear, pathway-specific projection and η_i contains biological variability and unmodeled state. Equation 8 is the formal sense in which a dendrite can store a 3D image: it stores a local transformation of a three-dimensional relational field.

$$\hat{\mathbf{X}}_{\tau|t}(\mathbf{r}) = \mathcal{R}[\mathbf{q}_t, \{\mathbf{m}_i\}, \{A_k\}, \mathbf{c}_t, \mathbf{B}_t]. \quad (9)$$

where $\mathbf{B}_t$ is current body and environmental state. The reconstructed state is generated now. It is not a read-only copy.

$$\hat{\mathbf{X}}_{\tau|t}(\mathbf{r}) = \mathcal{G}\left[\sum_i \alpha_i(t) \psi_i(\mathbf{r}; \mathbf{m}_i)\right]. \quad (10)$$

where each learned dendritic state parameterizes a relational basis function ψ_i , $\alpha_i(t)$ is current recruitment, and $\mathcal{G}$ is recurrent nonlinear integration. One dendrite can contribute a structured 3D component; the network composes the active image.

$$\{\mathbf{m}_i\} + \mathbf{q}_t \rightarrow S_k \rightarrow A_k \rightarrow \hat{\mathbf{X}}_{\tau|t}. \quad (11)$$

7. Tonic canvas and phasic ink

The **tonic canvas** is the ongoing receiver context against which a difference becomes consequential. It includes membrane potential, oscillatory phase, inhibition, neuromodulation, metabolic state, body state, and active recurrent relations. It is not a passive screen.

Phasic ink is the differentiated update that changes receivers and what they transmit. Depending on scale, it can include synaptic events, dendritic plateaus, spikes, bursts, or population departures. It is not identical to gamma activity in every circuit.

$$\mathbf{C}_{k+1} = \mathcal{T}(\mathbf{C}_k, \mathbf{P}_k, \mathbf{M}, \mathbf{I}_k, \mathbf{A}_k, \mathbf{O}_k). \quad (12)$$

The system performing Equation 12 is the observer-action process. No hidden agent watches the reconstruction.

8. Relationship to HTM and Thousand Brains

HTM and this SAN proposal share active dendritic pattern recognition, predictive depolarization, earlier firing of prepared cells, fast local inhibition, sparse population activity, and local learning. Hawkins and Ahmad (2016) explicitly describe previously depolarized cells firing sooner and fast inhibition biasing sequence-memory activation toward prediction. Ahmad and Hawkins (2016) separately formalize the accuracy and capacity of active dendritic recognition over sparse distributed representations.

Thousand Brains adds sensorimotor location, object-centered reference frames, and agreement among many cortical models (Hawkins et al., 2019; Lewis et al., 2019). These are proposed computational mechanisms, not established neocortical grid-cell facts, and they are close to SAN’s distributed reconstruction problem.

SAN adds six commitments:

1. oscillatory opportunity windows organize competition;
2. a dendritic disposition can represent a three-dimensional network-neighborhood projection;
3. selected local states telescope into distributed NAPOT reconstruction;
4. tonic and phasic activity have receiver-relative canvas and update roles;

5. perception and memory terminate in embodied observer-action rather than an inner decoder;
6. the mechanism is not assumed to be identical in every cortical layer, region, or modality.

The HTM Spatial Pooler remains an engineering comparator. Fixed sparsity percentages, instantaneous blanket inhibition, and universal minicolumn operations should not be presented as measured cortical facts without circuit-specific evidence.

9. Inversphere and the 3D-to-2D-to-3D intuition

$$\mathbf{y} = \mathbf{c} + \frac{R^2}{\|\mathbf{x} - \mathbf{c}\|^2}(\mathbf{x} - \mathbf{c}). \quad (13)$$

It preserves local angles while distorting scale. The visual effect can fold an extended environment around a sphere; conformal geometry supplies the mathematical analogy (Crane, 2013).

The analogy is useful because a large relational neighborhood can be expressed in a compact transformed structure and later expanded. It does not imply that dendrites execute Equation 13, contain Gaussian splats, or implement literal conformal geometry.

Micah Blumberg’s lucid 3D-to-2D-to-3D image can be interpreted as an analysis-synthesis cycle. Distributed sensory relations contract into learned projections. Recurrent inference and action expand those projections into a current 3D model. Some relations may be preserved while metric scale and detail change.

10. Falsifiable predictions

10.1 Dendritic geometry carries recoverable neighborhood information

After learning, dendritic and synaptic state should predict relational properties of the prior network neighborhood beyond current firing rate and local stimulus identity. A decoder should recover more than a category label: it should recover structured relations among participating pathways or positions.

10.2 Dendritic readiness predicts relative spike timing

Measured subthreshold or dendritic state should predict which cells fire earliest within a declared opportunity window beyond feedforward strength.

10.3 Early winners recruit identifiable inhibition

Perturbing the inhibitory route recruited by the early winner should broaden the selected set, alter temporal precision, and change later reinstatement.

10.4 Local selection predicts distributed 3D reconstruction

$$\rho_{\tau,t} = \text{sim}(\hat{\mathbf{X}}_{\tau|t}, \mathbf{X}_{\tau}). \quad (14)$$

Similarity must be reported separately for rate, timing, phase, connectivity, represented geometry, behavior, and subjective report where appropriate.

10.5 Timing matters beyond spike count

Holding approximate spike count constant while shifting phase or inhibitory timing should change the selected ensemble, reconstructed geometry, and behavior if oscillatory WTM is load-bearing.

10.6 Reconstruction is context-conditioned

The same local trace and cue should produce systematically different reconstructions under different tonic states, body positions, goals, or actions. This distinguishes regeneration from fixed playback.

10.7 Cross-scale perturbation and rescue

Disrupting a candidate dendritic projection should impair early selection and downstream reconstruction. Restoring the local disposition or its predicted timing should rescue both network geometry and behavior. Rescue of firing without reconstruction would count against the full mechanism.

11. Mathematical, digital, and biological proof program

11.1 Constructive representational theorem

Let an admissible three-dimensional scene satisfy $\mathbf{x} = \mathbf{B}\mathbf{z}$, where $\mathbf{B}$ spans a declared k -dimensional spatial model. Let learned dendritic state store measurements $\mathbf{y} = \mathbf{D}\mathbf{x} + \mathbf{e}$, and define $\mathbf{M} = \mathbf{D}\mathbf{B}$. If $\mathbf{M}$ has full column rank and $\mathbf{e} = 0$, then

$$\hat{\mathbf{x}} = \mathbf{B}\mathbf{M}^\dagger \mathbf{y} = \mathbf{x}. \quad (15)$$

This proves that a dendritic measurement state can store sufficient information to reconstruct every 3D image in the declared model family. It does not imply that finitely many scalar branch states can losslessly store an arbitrary unconstrained voxel field. Capacity depends on the dimensionality and structure of the admissible scene family.

With additive storage or retrieval noise, reconstruction obeys

$$\|\hat{\mathbf{x}} - \mathbf{x}\|_2 \leq \|\mathbf{B}\mathbf{M}^\dagger\|_2 \|\mathbf{e}\|_2. \quad (16)$$

The smallest singular value and conditioning of $\mathbf{M}$ therefore supply falsifiable robustness predictions. The constructive derivation and executable simulation are provided in [the proof-simulation package](#).

11.2 Digital reconstruction test

The reference simulation stores a structured 1,728-voxel volume through 24 simulated dendritic measurements over a six-dimensional spatial basis. The effective operator had rank 6 and smallest singular value 0.05744. Noiseless relative reconstruction error was $9.71\text{e-}16$. Under additive noise, error was 0.03280 against the proved upper bound 0.18765. A partial cue selected perspective 3 from eight stored candidates with winner margin 0.30592. Exact recovery persisted after reducing the retained measurements from 24 to 6 while full rank remained.

These results verify the declared linear construction, the implementation, and the dependence on rank. They do not demonstrate a learned decoder, biological storage, oscillatory selection, or recurrent NAPOT dynamics. The pseudoinverse is supplied directly. A stronger simulation must replace it with a learned recurrent excitatory-inhibitory network, impose conductance and timing constraints, and test whether iterative redistribution converges to the captured viewpoint without access to the analytic inverse.

11.3 Learned recurrent excitatory-inhibitory reconstruction

The analytic construction establishes a ceiling but does not show that a finite iterative network can approach it. A second deterministic fixture therefore generated the same six-dimensional family and 24 branch measurements, learned the latent-to-branch operator from 2,400 noisy training examples with a delta rule, and inferred 384 held-out latent states through a stable recurrence:

$$\mathbf{z}_{t+1} = \mathbf{W}_E \mathbf{z}_t - \mathbf{W}_I \mathbf{z}_t + \eta \hat{\mathbf{A}}^\top \mathbf{y}, \quad (17)$$

where

$$\begin{aligned} \mathbf{W}_E &= \left[\mathbf{I} - \eta \left(\hat{\mathbf{A}}^\top \hat{\mathbf{A}} + \lambda \mathbf{I} \right) \right]_+, \\ \mathbf{W}_I &= \left[-\mathbf{I} + \eta \left(\hat{\mathbf{A}}^\top \hat{\mathbf{A}} + \lambda \mathbf{I} \right) \right]_+. \end{aligned} \quad (18)$$

This is an algebraic excitatory-minus-inhibitory decomposition of a signed latent recurrence, not a claim that each matrix row is a Dale-constrained biological neuron. The learned operator had relative error 0.001648 and the recurrent matrix had spectral radius 0.866787. After 180 iterations, the median held-out reconstruction error was 0.056121 and the 90th percentile was 0.106308. One iteration produced median error 0.656311; shuffled measurement-scene pairings produced 1.407463. The true-operator pseudoinverse ceiling was 0.056281 and was not used during recurrent inference.

The recurrence remained stable and full rank after retaining 20, 16, 12, 8, or 6 measurements. Median error rose from 0.063045 at 20 retained measurements to 0.179391 at 6. This graded degradation is preferable to an all-or-none demonstration, but it still reflects a low-dimensional Gaussian-basis fixture whose operator family was represented in training.

Claim ceiling. The result shows that a learned forward operator and stable recurrent difference dynamics can reconstruct this declared finite-dimensional family without applying a pseudoinverse at inference. It does not show that dendrites store voxel images, that cortical inhibition has these weights, that the brain knows the forward operator, or that the reconstructed state is conscious. A biologically stronger next model must impose local learning, Dale-constrained cells, bounded firing states, realistic conductances, route delays, and matched recurrent alternatives.

11.4 Dale-sign-constrained delayed recurrence: frozen adverse-result test

The Draft 3 recurrence was an algebraic signed decomposition, not a cell-level biological model. Before inspecting the next result, the package froze a stronger question and six conjunctive support conditions. The new fixture represented six signed latent dimensions with 12 nonnegative push-pull channels, stored eight sparse prototypes, learned from 3,200 noisy examples, and tested 640 new examples with approximately 30% branch-measurement dropout. Twelve excitatory and three inhibitory units obeyed fixed output signs; all states were clipped to $[0, 1]$; excitatory and inhibitory conductance terms used two-step delayed activity over 90 iterations. Input weights used a projected pre-activity times postsynaptic-error rule, while recurrent excitation used a projected Hebbian / Oja-like coactivation rule. The constrained model used no pseudoinverse.

The predeclared support rule required the constrained recurrence to pass every comparison, not merely remain stable:

Frozen condition	Observed	Required	Result
Median-error gain over matched feedforward	0.026893	at least 0.02	pass
Prototype-accuracy gain over matched feedforward	-0.003125	at least 0.03	fail
Median-error gain over no-inhibition ablation	0.002280	at least 0.01	fail
Median-error gain over shuffled pairing	0.509918	at least 0.10	pass
Dale sign and bounded-state checks	0 violations; bounded and finite	zero; bounded and finite	pass

The complete endpoint table is:

Model	Median relative error	90th-percentile error	Prototype accuracy
Dale-constrained delayed recurrence	0.491514	0.741693	0.859375
Matched feedforward	0.518407	0.753649	0.862500
No-inhibition ablation	0.493794	0.743223	0.853125
Shuffled measurement-target pairing	1.001433	1.158176	0.114062
Signed ridge baseline	0.374549	0.556704	0.993750

The constrained network clearly preserved real measurement-target information: it strongly exceeded shuffled pairing. It also remained finite, bounded, and Dale-sign-consistent. Those facts do not rescue the primary hypothesis. Prototype accuracy was 0.003125 lower than matched feedforward, the error gain over the no-inhibition ablation was only 0.002280, and signed ridge decoding was substantially better on both endpoints. Under the frozen conjunctive rule, `primary_hypothesis_supported` is false.

Interpretive consequence. This fixture does not establish that delayed recurrence or inhibition supplied a winner-take-most reconstruction advantage. The learned input map and push-pull representation account for most of the observed completion in this task. The biological proposal remains open, but a future positive test must earn its recurrent and inhibitory terms against capacity-matched controls rather than infer mechanism from successful reconstruction alone. Reasonable next tests include spiking conductance dynamics, cell-type-specific inhibitory motifs, oscillatory phase perturbations, separately learned out-of-family scenes, and perturbation-rescue experiments with preregistered endpoints. The failed result must remain part of the evidence record even if later models succeed.

11.4.1 Frozen 12-seed sensitivity audit

A separate specification was frozen before varying the generator seed. Twelve consecutive seeds (20260721-20260732) regenerated the prototypes, samples, branch operator, noise, dropout, initial weights, and shuffled pairing while preserving all architecture, hyperparameter, endpoint, comparator, and threshold choices. This audit cannot overturn the original failed primary result; it asks whether that failure was narrowly seed-specific.

Frozen condition	Seeds passing / 12
Median-error gain over matched feedforward	6
Prototype-accuracy gain over matched feedforward	0

Frozen condition	Seeds passing / 12
Median-error gain over no-inhibition ablation	0
Median-error gain over shuffled pairing	12
Sign and bounded-state checks	12
Full conjunctive primary rule	0

Across seeds, the constrained model’s median-of-seed median relative error was 0.468764 (range 0.365544-0.559576), compared with 0.478934 for matched feedforward, 0.470602 for no inhibition, and 0.359779 for signed ridge. Its median-of-seed prototype accuracy was 0.827344, versus 0.822656 feedforward and 0.988281 ridge. Zero seeds passed the full rule. The stable result is therefore not that the fixture failed to learn: every seed beat shuffled pairing. The stable adverse result is that the tested delayed recurrence and inhibitory conductance did not earn the declared incremental advantage.

11.4.2 What the constrained fixture does not test

Four method boundaries are load-bearing:

1. **Supervision.** The input update is presynaptic activity multiplied by a target-derived postsynaptic error and then projected nonnegative. It is locally factorizable only if an appropriate cell-level teaching signal exists; the fixture does not supply or validate that signal.
2. **Stored-family completion.** Held-out samples contain new noise and branch dropout but remain variations of the same eight stored prototypes under the same operator. This is associative completion, not transfer to unseen prototype families or operators.
3. **External readout.** Prototype accuracy is computed after latent decoding by nearest-prototype distance. It is not an internally observed winner whose phase, recruitment, and inhibitory suppression instantiate oscillatory WTM.
4. **Latent rather than voxel reconstruction.** The constrained fixture estimates six latent coordinates. The separate analytic construction maps a six-dimensional basis into a 1,728-voxel volume; the constrained network does not learn or directly evaluate that spatial synthesis.

The constrained architecture reports 792 weights (576 input, 144 recurrent excitatory, 36 excitatory-to-inhibitory, and 36 inhibitory-to-excitatory), compared with 288 signed-ridge weights. Parameter count is not a complete cross-architecture capacity measure, but the smaller ridge model’s better performance blocks any claim that the constrained fixture is currently more parsimonious or efficient. The sign restrictions and dynamics remain scientifically relevant constraints; they are not evidence of biological identity.

11.4.3 Why no post hoc positive test is added

Two further tests were considered after the adverse result: transfer to unseen prototype or operator families, and an internally observed winner. The present fixture cannot answer either question without changing its data-generating process, architecture, endpoint, and support rule after seeing the failure. Its nearest-prototype label is external, and its rate dynamics contain no declared phase-defined internal winner. Adding a favorable endpoint now would be post hoc and scientifically weak.

The next model should therefore be preregistered separately. It must hold out complete prototype families and measurement operators, define winner identity from network activity before decoding, manipulate inhibitory timing causally, and compare phase-sensitive selection against capacity-matched feedforward, no-inhibition, recurrent-attractor, and signed linear controls. This final preprint treats those requirements as future discriminating work and draws no present support from them.

11.5 Synapse-count and image-capacity bounds

The storage question changes substantially when the dendritic state is allowed to include thousands of synapses rather than a few scalar branch measurements. The relevant anatomical counts are cell-type specific. Detailed models of human temporal-cortex layer 2/3 pyramidal neurons estimated approximately 30,000 excitatory synapses per cell (Eyal et al., 2018). A quantitative anatomical study of rat cerebellar Purkinje cells estimated approximately 175,000 parallel-fiber synapses per dendritic tree, with a separate varicosity-based route near 2.09×10^5 (Napper and Harvey, 1988). The latter is evidence that some neurons receive on the order of 200,000 synaptic contacts; it is not a count that should be assigned to a generic cortical pyramidal neuron.

Let synapse s possess Q_s reliably distinguishable long-lived states. Ignoring correlations and shared constraints, the gross synaptic-state ceiling is

$$C_{\text{gross}} \leq \sum_{s=1}^S \log_2 Q_s \quad \text{bits.} \quad (19)$$

A 2024 information-theoretic analysis of CA1 dendritic spine-head volumes estimated 24 distinguishable sizes and approximately 4.1-4.59 bits per synapse (Samavat et al., 2024). Earlier nanoconnectomic work reported approximately 4.7 bits under a related method and noted the averaging required by stochastic activation (Bartol et al., 2015). These estimates concern particular hippocampal spine datasets and averaging conditions; they are not established universal capacities for all synapses, nor proof that every distinguishable morphological state is independently available to NAPOT.

Using 4.1-4.59 bits only as an illustrative range gives:

Illustrative neuron	Synapses	Gross state ceiling	Binary-voxel cube with the same bit count
Human L2/3 pyramidal-cell model	30,000	123,000-137,700 bits (15.0-16.8 KiB)	approximately 50^3 to 52^3
Rat Purkinje-cell anatomical estimate	175,000	717,500-803,250 bits (87.6-98.1 KiB)	approximately 90^3 to 93^3
Round high-convergence example	200,000	820,000-918,000 bits (100.1-112.1 KiB)	approximately 94^3 to 97^3

The voxel comparison is deliberately crude. A synapse is not a voxel, a 3D scene is not generally an uncompressed binary occupancy grid, and synaptic states are correlated, noisy, metabolically constrained, and continuously remodeled. Conversely, a learned generative basis can reconstruct a much larger apparent scene from a lower-dimensional code because the network supplies regularities, priors, current sensory evidence, body position, and recurrent completion.

For the linear model, the recoverable scene dimension obeys

$$k_{\text{recoverable}} \leq \text{rank}(\mathbf{DB}) \leq \min(S_{\text{eff}}, k_{\text{model}}), \quad (20)$$

where S_{eff} is the number of effectively independent synaptic or dendritic measurement dimensions, not the raw anatomical synapse count. If each reconstructed voxel requires b_v independent bits, a gross cubic-resolution ceiling is

$$L_{\max} \leq \left(\frac{C_{\text{effective}}}{b_v} \right)^{1/3}. \quad (21)$$

The central empirical unknown is therefore not merely S . It is the effective capacity $C_{\text{effective}}$ after accounting for correlated inputs, branch nonlinearities, turnover, coding precision, cue selectivity, decoder accessibility, and error correction. SAN / NAPOT predicts that reconstruction quality should scale more strongly with this effective rank and branch organization than with synapse count alone.

11.6 Biological identification

A decisive experiment should record:

- identified dendritic and subthreshold states;
- local excitatory and inhibitory populations;
- distributed sensory, hippocampal, association, and motor activity;
- body position, action, and outcome.

An animal could learn overlapping three-dimensional sensorimotor episodes. A partial cue would leave multiple candidate reconstructions. Two-photon voltage or calcium imaging, targeted dendritic recording, high-density electrophysiology, and cell-type-specific optogenetics would estimate Equations 2-10.

Critical manipulations would:

1. alter a learned dendritic segment without changing the cue;
2. delay or weaken recruited inhibition;
3. preserve firing rate while shifting spike phase;
4. activate the putative winner before, within, or after the natural window;
5. test whether restoring timing rescues distributed geometry and behavior.

The analysis must compare capacity-matched models:

- feedforward rate plus generic inhibition;
- HTM-like predictive depolarization;
- recurrent attractor or hippocampal pattern completion;
- generic recurrent neural networks;
- SAN telescoping with dendritic 3D projection, tonic context, phase-sensitive selection, and NAPOT reconstruction.

12. Alternative explanations and evidence boundary

Feedforward competition, divisive normalization, recurrent attractors, hippocampal indexing, predictive coding, and generic recurrent networks can explain parts of the proposed behavior. SAN is supported only if its declared dendritic projection, timing, inhibition, and multiscale reconstruction variables provide held-out causal value beyond those alternatives.

Dendritic nonlinearity alone is insufficient. Sparse activity is insufficient. Oscillatory correlation is insufficient. Single-cell behavioral influence is insufficient. Successful HTM simulation is insufficient. The hypothesis requires the full chain from learned local 3D projection through timed excitatory-inhibitory selection to distributed reconstruction and action. The constrained synthetic test and its 12-seed sensitivity audit further show that stable Dale-sign-consistent recurrence, bounded activity, and successful associative decoding are also insufficient: the recurrent and inhibitory terms must outperform matched alternatives under

frozen criteria. The current nearest-prototype endpoint cannot by itself identify an internal winner-take-most mechanism.

13. Implications

If supported, the model would connect memory, perception, and choice without a central executive. A local dendritic projection would not decide by itself. It would change which larger state becomes reachable. That state changes available evidence, prepared actions, and outcomes feeding the next cycle.

The proposal also gives SAN’s camera, screen, receiver, and transmitter language a biological interpretation. Dendrites receive and structurally differentiate input relations. Axons and terminals transmit changes that reshape downstream fields. A dendrite may store a 3D relational image in its learned physical organization, but neither dendrite nor neuron is a little viewer. The active network reconstruction is the seeing, remembering, and choosing process.

14. Conclusion

Telescoping network reinstatement proposes that a dendritic arbor can store a learned three-dimensional projection of its prior network neighborhood and later seed regeneration of that state through NAPOT. Nonlinear dendritic integration supplies structured readiness. Oscillatory excitation-inhibition creates selection windows. Earlier firing recruits inhibition and biases winner-take-most assemblies. Recurrent pathways expand the selected state into a context-sensitive 3D model that influences perception, memory, choice, and action.

HTM provides a computational predecessor for predictive depolarization, earlier firing, local inhibition, and sparse sequence memory. SAN’s extension is the link to 3D dendritic projection, tonic and phasic receiver roles, multiscale NAPOT reconstruction, and distributed observer-action.

The proposal remains unconfirmed. A sentence-level audit of all 18 references supports the component claims and comparators but supplies no validation of the joined SAN mechanism. The first two constructions establish finite-dimensional representational and iterative sufficiency, while the Dale-sign-constrained rate fixture fails its frozen primary support rule in the original seed and all 12 sensitivity seeds. That robust synthetic failure narrows the current evidence: supervised learned input structure can support same-family partial-cue reconstruction, but delayed recurrence and inhibition have not yet earned a winner-take-most causal interpretation. The paper’s value is that it converts the image of a world folding into a dendrite and unfolding into a brain state into a mechanism that can be measured, compared, perturbed, rescued, or falsified, including by adverse results.

Authorial provenance and research chronology

Artificial intelligence was not involved in authoring or originating the core ideas, scientific theories, hypotheses, or source-based research presented in this paper. The underlying research program was not written or developed by AI. This body of work is Micah Blumberg’s original research, developed through a continuous program spanning 2011-2026. Its central conceptual foundations were developed and recorded before ChatGPT’s public release on November 30, 2022.

In later manuscript development, AI tools were used only in bounded supporting roles, including assistance with mathematical and computational formulation, code, grammar and spelling, and checks against medical, biological, and physics literature. These tools did not supply the original theories or authorship. Micah

Blumberg selected the claims and sources, directed the analysis, approved the final language, and remains responsible for the paper’s arguments, interpretations, and errors.

Data and code availability

The public release bundle includes the constructive proof, executable simulations, frozen specifications, and deterministic machine-readable results in supplemental-computation/. The learned recurrent fixture is supplemental-computation/simulate_learned_recurrent_ei_reconstruction.py; its result is supplemental-computation/learned-recurrent-ei-results.json. The frozen constrained specification, implementation, and adverse result are supplemental-computation/CONSTRAINED-RECURRENT-SPEC.md, supplemental-computation/simulate_constrained_dale_reconstruction.py, and supplemental-computation/constrained-dale-reconstruction-results.json. The frozen sensitivity specification, implementation, and 12-seed result are likewise included. All simulations use generated data and perform no human or animal experiment. Exact source and result hashes are recorded in PUBLIC-MANIFEST.sha256; the public revision and evidence boundary is summarized in REVISION-AND-CLAIM-SCOPE.md. Manuscript status alone does not authorize a repository release.

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